

Employee Attrition Prediction

Predicting resignations to enable proactive HR retention

IBM HR Analytics dataset • EDA → Feature Engineering → Modeling → SHAP
→ ROI

\$22.5M

Annual attrition cost

0.83

Best model ROC-AUC

81%

Leavers detected (recall)

\$4.5M

Projected savings

Prepared as an end-to-end, interview-grade HR analytics case study. All figures are reproduced from code in this repository.

1. Business Problem

A company with **10,000 employees** loses **15%** of its workforce every year — about **1,500 people**. Each departure costs roughly **\$15,000** (\$5,000 recruitment + \$3,000 training + \$7,000 lost productivity), for a total annual loss of **\$22.5 million**.

Objective. Predict which employees are likely to resign in the next 3–6 months so HR can intervene proactively — before the resignation, not after.

Dataset. IBM HR Analytics — 1,470 employees, 35 raw attributes spanning demographics, compensation, job role and satisfaction scores. Target: **Attrition** (Yes/No). The positive class is the minority at 16.1%.

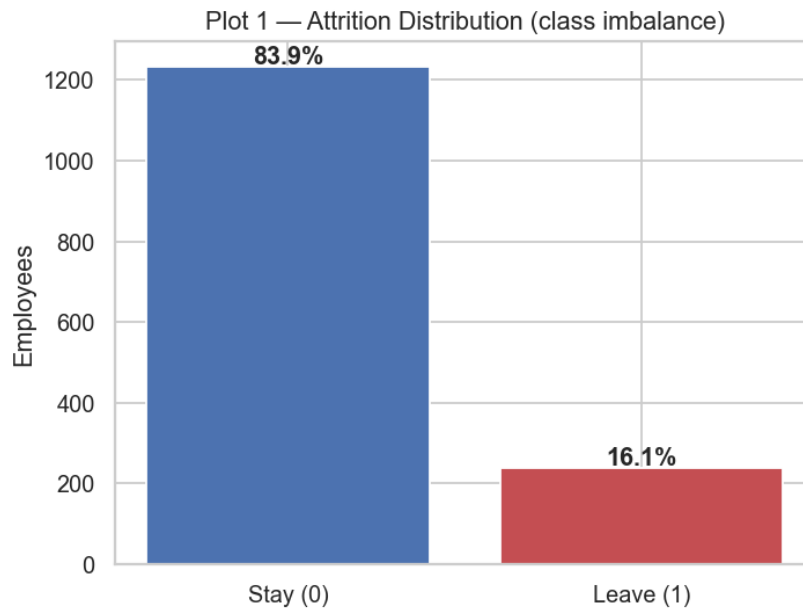
2. Initial Business Questions & EDA Findings

Six questions were answered directly from the data before any modeling. Each finding maps to an HR action.

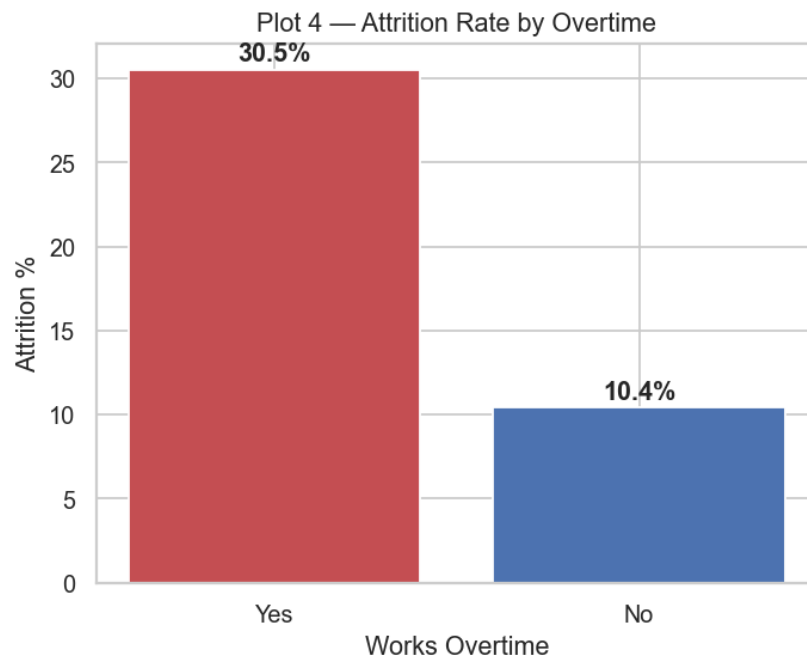
Question	Data-driven answer
Which departments lose most employees?	Sales has the highest attrition at 20.6%, vs Research & Development at 13.8%.
Does overtime cause resignations?	Overtime employees leave at 30.5% vs 10.4% without overtime (2.9x higher).
Do low salaries increase churn?	Median income of leavers is \$3,202 vs \$5,204 for stayers (38% lower).
Are new employees leaving quickly?	Employees with <2 years tenure leave at 34.9% vs 12.9% for more tenured staff.
Does promotion delay impact attrition?	Employees not promoted in >4 years leave at 14.2% vs 16.5% for recently promoted staff.
Which employee profiles are most risky?	Highest risk profile: works overtime, in the lowest salary band, early career (<2y), with delayed promotion and low job satisfaction (satisfaction=1 leaves at 22.8%).

Note: in this dataset promotion delay alone is *not* a strong univariate driver — recently promoted staff churn at a similar rate. It matters only in combination with other factors, which is exactly what the models and SHAP capture.

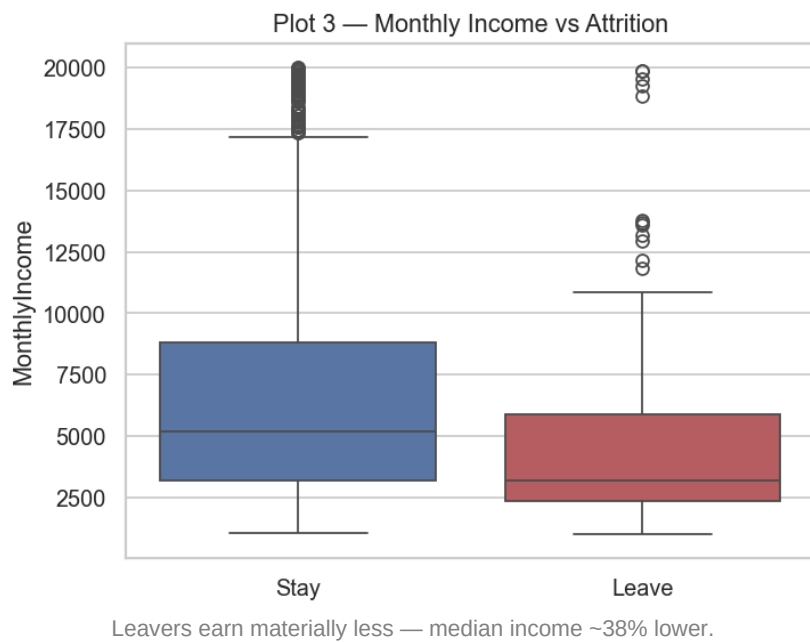
Key EDA Visuals



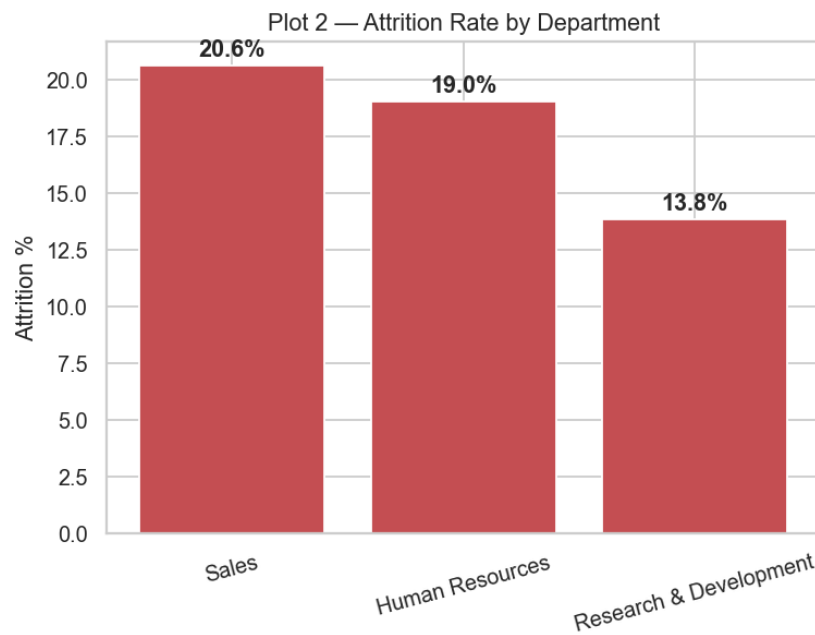
Class imbalance — leavers are the minority, so we track recall/precision, not accuracy.



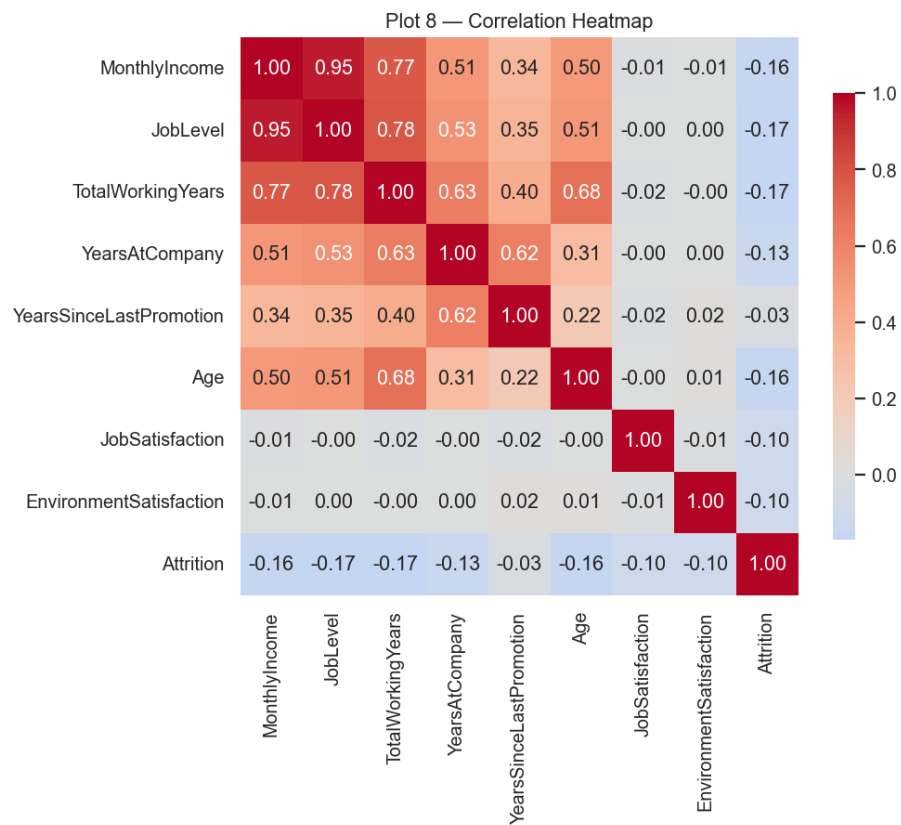
Overtime is the single starkest split — overtime staff leave ~3x more often.



Resignations cluster in the first 1–3 years — an onboarding/early-tenure problem.



Sales has the highest departmental attrition.



MonthlyIncome, JobLevel and TotalWorkingYears are highly correlated (redundancy to watch).

3. Feature Engineering

Raw fields were enriched with business-driven features, each encoding a testable HR hypothesis:

Engineered feature	Business meaning
SalaryBand / SalaryPercentile	Internal pay fairness (terciles + percentile rank)
ExperienceGroup	Attrition behaviour differs: Fresh / Mid / Senior
PromotionDelayFlag	Career stagnation (>4 years since promotion)
LongCommuteFlag / CommuteCategory	Travel fatigue (distance > 20)
OverTimeFlag	Burnout indicator
EarlyCareerFlag	Weak onboarding / early churn (<2 years)
TrainingGapFlag	Lack of development (0 trainings last year)
IncomeVsLevelRatio	Pay relative to peers at the same job level

Constant columns (EmployeeCount, StandardHours, Over18) and the EmployeeNumber identifier were dropped. Categoricals were one-hot encoded; numerics standardised. The split is stratified 80/20 to preserve the attrition ratio in both sets.

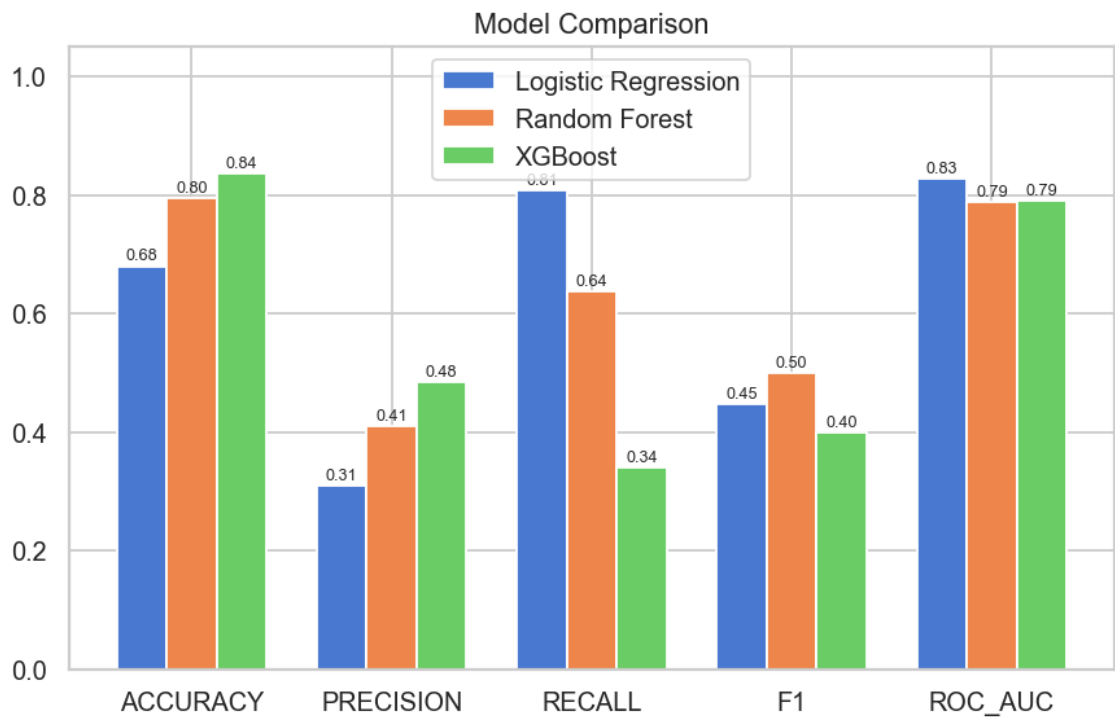
4. Models & Evaluation

Three classifiers were trained with class-imbalance handling (balanced class weights / scale_pos_weight) and tuned with 5-fold randomised search optimising ROC-AUC. Because the positive class is rare, we evaluate on precision, recall, F1 and ROC-AUC — not raw accuracy.

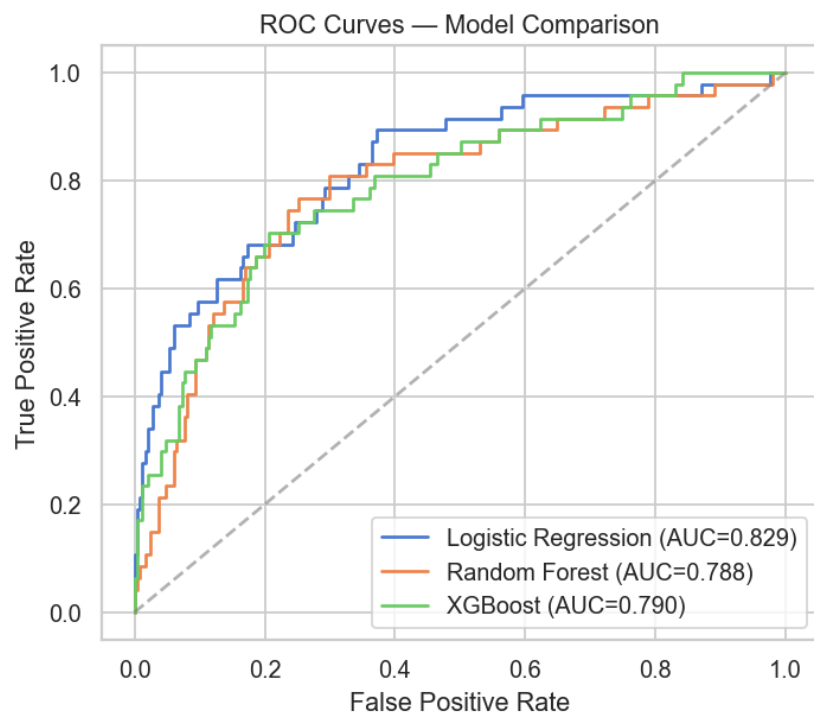
Model	Accuracy	Precision	Recall	F1	ROC-AUC
Logistic Regression	0.68	0.31	0.81	0.45	0.83
Random Forest	0.80	0.41	0.64	0.50	0.79
XGBoost	0.84	0.48	0.34	0.40	0.79

Winner: Logistic Regression — selected on ROC-AUC (0.83, threshold-independent) with the best recall (81%). A notable, honest result: on this small (~1.5k row), largely linear dataset a well-regularised logistic regression with balanced class weights *beat* tuned Random Forest and XGBoost. Complexity is not automatically better — model choice must be evidenced, not assumed.

Decision threshold lowered to 0.35 so the business catches more true leavers (higher recall) at the cost of some false alarms — the right trade-off when a missed leaver costs \$15k and a false alarm costs a retention conversation.



Side-by-side metric comparison.

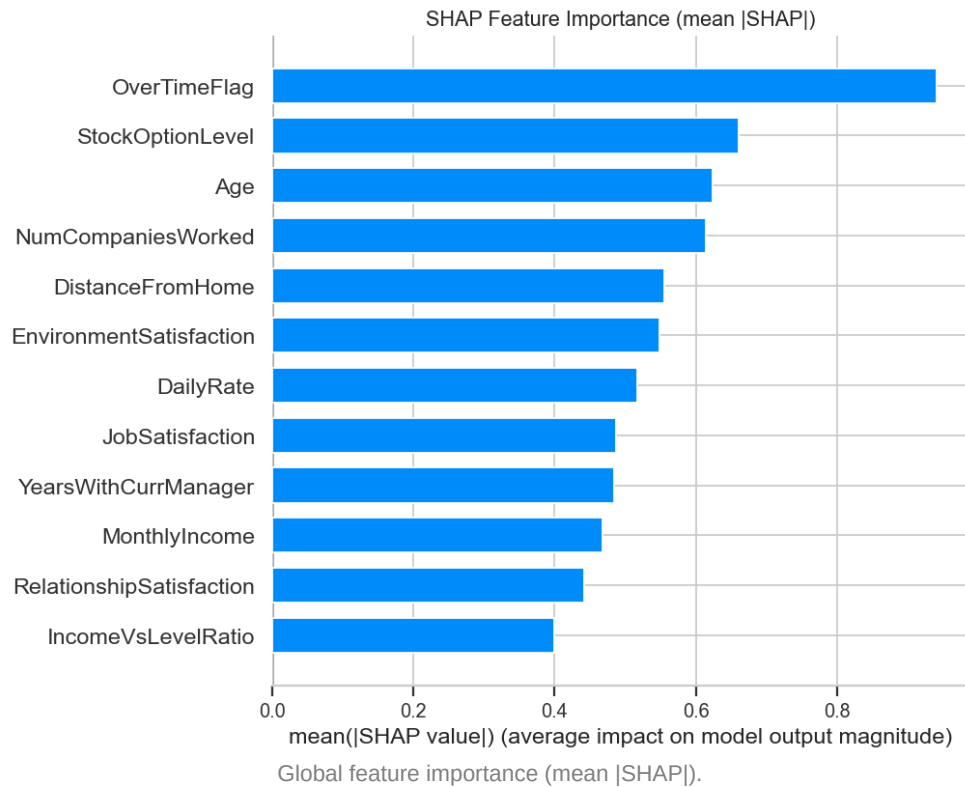


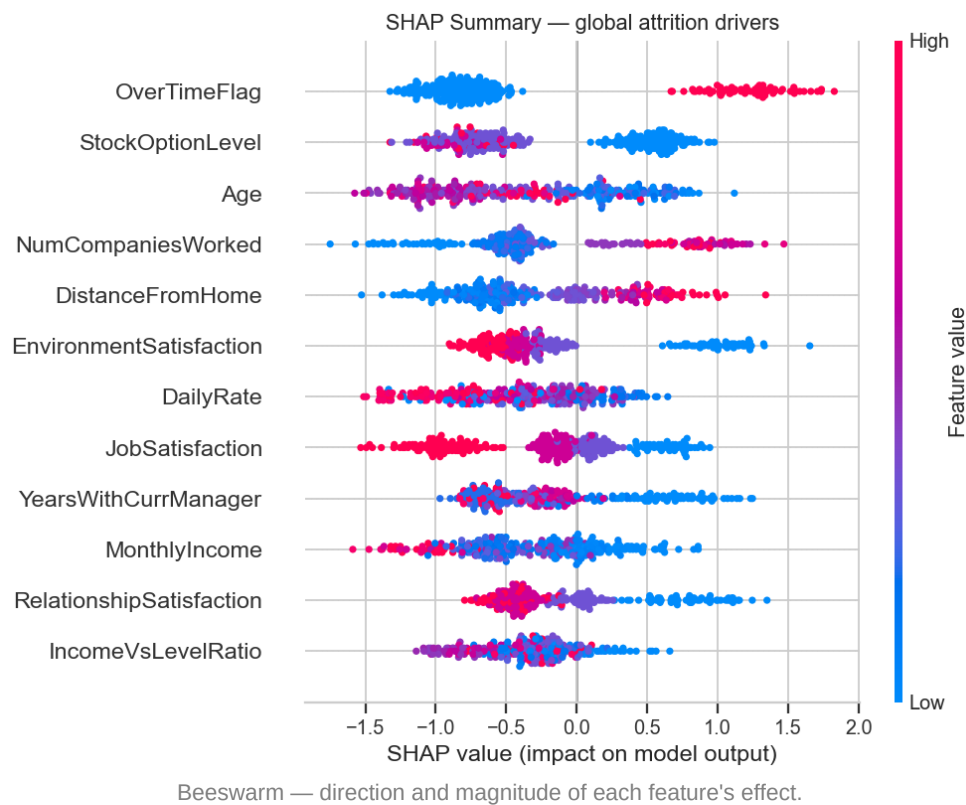
ROC curves — ranking quality across models.

5. Explainability with SHAP

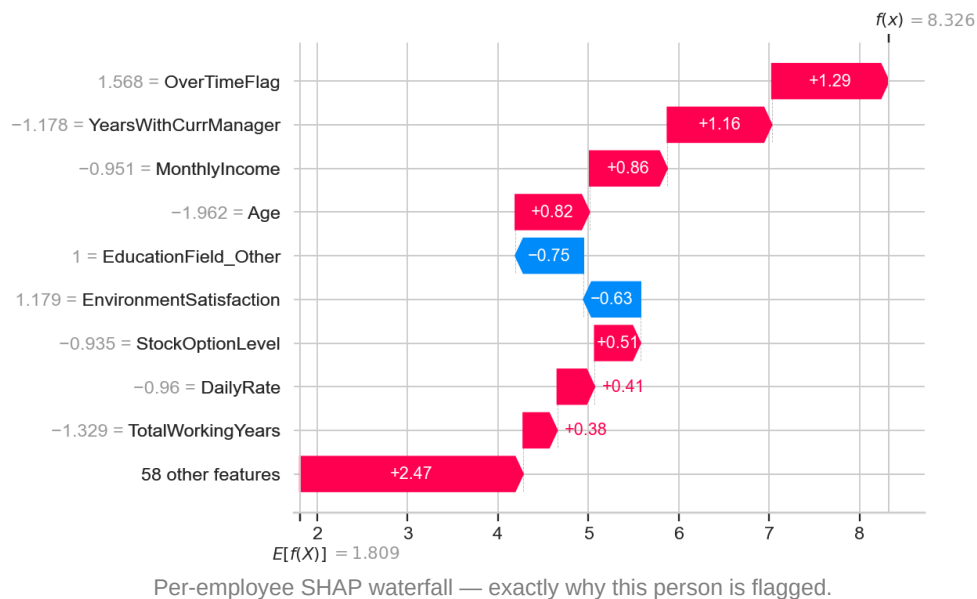
HR will always ask “*why is this person predicted to leave?*”. SHAP decomposes every prediction into per-feature contributions, giving both a global driver ranking and a per-employee explanation.

Top global drivers: OverTimeFlag, StockOptionLevel, Age, NumCompaniesWorked, DistanceFromHome, EnvironmentSatisfaction.





Worked example. The highest-risk employee in the test set was predicted at **100% risk** and did in fact leave (actual = 1). The model's reasons: OverTimeFlag, YearsWithCurrManager, MonthlyIncome, Age, StockOptionLevel. HR gets not just a score but the *levers to pull*.



6. Business Recommendations

Recommendation	Evidence	Potential impact
Reduce overtime / rebalance workload	Strongest driver in both EDA and SHAP	Attrition ↓ 10–15%
Salary correction for bottom 20%	Leavers earn ~38% less; pay is a top driver	Attrition ↓ 5–10%
Structured onboarding for first 2 years	Churn peaks in years 1–3	Attrition ↓ 8–12%
Promotion / career-path review	Stagnation compounds with other risks	Attrition ↓ ~5%
Manager & satisfaction interventions	Low satisfaction + manager tenure are drivers	Attrition ↓ ~5%

7. Final Business Value (ROI)

Of 1,500 annual leavers, the model surfaces ~**81%** (measured test-set recall). Assuming HR successfully retains **25%** of those flagged, that saves **303 employees** per year:

\$22.5M Annual loss today	303 Employees saved / yr	\$15,000 Cost per leaver	\$4.5M Annual savings
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Calculation: 1,500 leavers × 81% detected × 25% retained = 303 employees × \$15,000 = **\$4,547,812 per year**.

Every assumption above is defined in `src/config.py` and the recall is measured, not assumed — so the ROI is fully auditable.

8. What Makes This Project Stand Out

- Frames a real business problem and quantifies it in dollars before any code.
- EDA answers concrete HR questions, not generic charts.
- Feature engineering is hypothesis-driven, each feature tied to an action.
- Three models compared with proper imbalance handling and tuning.
- Honest model selection — reports that the simplest model won, with reasoning.
- SHAP gives per-employee explanations HR can act on.
- Closes the loop with recommendations and an auditable multi-million-dollar ROI.